AI-Based Emergency Call and SMS Triage System for Sri Lankan Disasters

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT

Sri Lanka is susceptible to natural disasters like floods, landslides and cyclones that cause sudden spikes in emergency hotline calls and text messages. Emergency operators at such times need to urgently extract clear information from callers in distress who may speak different languages including Sinhala, Tamil and English. In a lot of cases, callers give ambiguous information under panic, noise or poor communication conditions. This leads to operators having to ask more clarification questions, wasting valuable time and delaying emergency response with increased risk of loss of human life.

Current emergency triage systems are mainly focused on high-resource languages and assume either structured call workflows or extensive manual questioning by operators. These systems are not designed for multilingual and code-mixed communication, ubiquitous in Sri Lanka, nor do they offer intelligent methods of autonomously collecting information that may be missing from emergency calls.

This research proposes an **AI BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW RESOURCE LOCAL LANGUAGES,** which includes an API-based valuable addition, a **Multilingual Emergency Conversational AI Agent for Rapid Clarification**. The proposed agent reviews information extracted from emergency calls and automatically recognizes missing critical information such as location, number of people affected, or severity of incidents. The agent then provides brief, targeted clarification questions in the caller's language to swiftly obtain the sought information.

This solution will leverage ASR, NLP, LLMs for multilingual language detection and machine learning techniques utilizing Python with HuggingFace Transformers and system architectures employing APIs

Expected results are faster information gathering, better accuracy of emergency triage and reduced workload of emergency operators. By using this system itself Socially, it can enhance public safety and efficiency in disaster response and commercially they have potential applications for emergency services, disaster management authorities and public safety organizations.

Keywords: Emergency AI, Multilingual NLP, Disaster Response Systems, Conversational AI, Emergency Call Triage.

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We would like to convey our heart feared thankfulness for all assistance and guidance received in preparing this research proposal **"AI BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW RESOURCE LOCAL LANGUAGES"**

We would like to take this opportunity at the outset to express our deepest gratitude to our supervisors **Dr. Prasanna Sumathipala**, our research supervisor, and **Ms. Malithi Nawarathne,** co-supervisor for their invaluable guidance, constructive criticism and encouragement throughout the development of this research proposal. Their expert advice, insightful suggestions, and support have been invaluable in guiding the focus and quality of our research.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

API – Application Programming Interface

ASR – Automatic Speech Recognition

JSON – JavaScript Object Notation

LLM – Large Language Model

ML – Machine Learning

NLG – Natural Language Generation

NLP – Natural Language Processing

NLU – Natural Language Understanding

WBS – Work Breakdown Structure

1. INTRODUCTION

1.1 Background and Context

Floods, landslides, cyclones and severe storms are natural disasters that affect many countries around the globe," including Sri Lanka. Such disasters usually result in a sudden increase in emergency call center activity as people look for immediate help from the emergency services. In such situations, emergency call operators need to quickly collect key information that includes the caller's location, the kind of incident, how many people it is affecting and what the level of danger was. But callers are often in emotional distress or panic, which can lead to uncommunicative behavior. Operators therefore struggle to collect accurate information fast which can slow emergency responses.

Emergency communication systems are challenged by practical difficulties like noisy environments, potential distress of callers, heterogeneity in language usage and pressure of time. In countries with multiple languages such as Sri Lanka, the callers can talk to us in Sinhala, Tamil or English; sometimes mixed words too - such as Singlish or Tanglish. This diversity adds to the challenge for emergency operators to efficiently and effectively hear, understand and process information. And when key details don't come across or are vague, operators need to take extra time asking follow-up questions - which can ultimately delay the dispatch of emergency services.

Currently, most emergency call centers are greatly dependent on human operators to manually obtain information from callers. Although this approach enables flexible interaction between operators and callers, it has also been observed that it increases the workload of operators during emergency situations when a huge volume of calls is received at the same time. Currently, most technological solutions are mainly focused on call routing or speech transcription; however, most of these solutions lack smart capabilities that can help operators obtain missing information from callers during emergency situations.

Therefore, considering the above challenges facing emergency call centers, this research aims to propose an innovative solution that will be referred to as an **'AI-Based Multilingual Emergency Call and SMS Triage System.'** The proposed system has been developed to help emergency operators during emergency situations by enabling them to better comprehend emergency situations. The proposed system has various intelligent components that can analyze emergency calls as well as SMS messages to better comprehend emergency situations. One of the components of the proposed system is referred to as a **'Multilingual Emergency AI Agent for Rapid Clarification'** which is able to identify missing information from callers.

This research is also consistent with some of the **United Nations Sustainable Development Goals**. These are **SDG 3:** Good Health and Well-being through better emergency response systems; **SDG 9:** Industry, Innovation, and Infrastructure using advanced artificial intelligence technologies; and **SDG 11:** Sustainable Cities and Communities through better disaster response.

The proposed system is under the research cluster of **Autonomous Intelligent Machines and Systems (AIMS)** as it is focused on creating a system that can utilize artificial intelligence to help human decision-making in real-world scenarios.

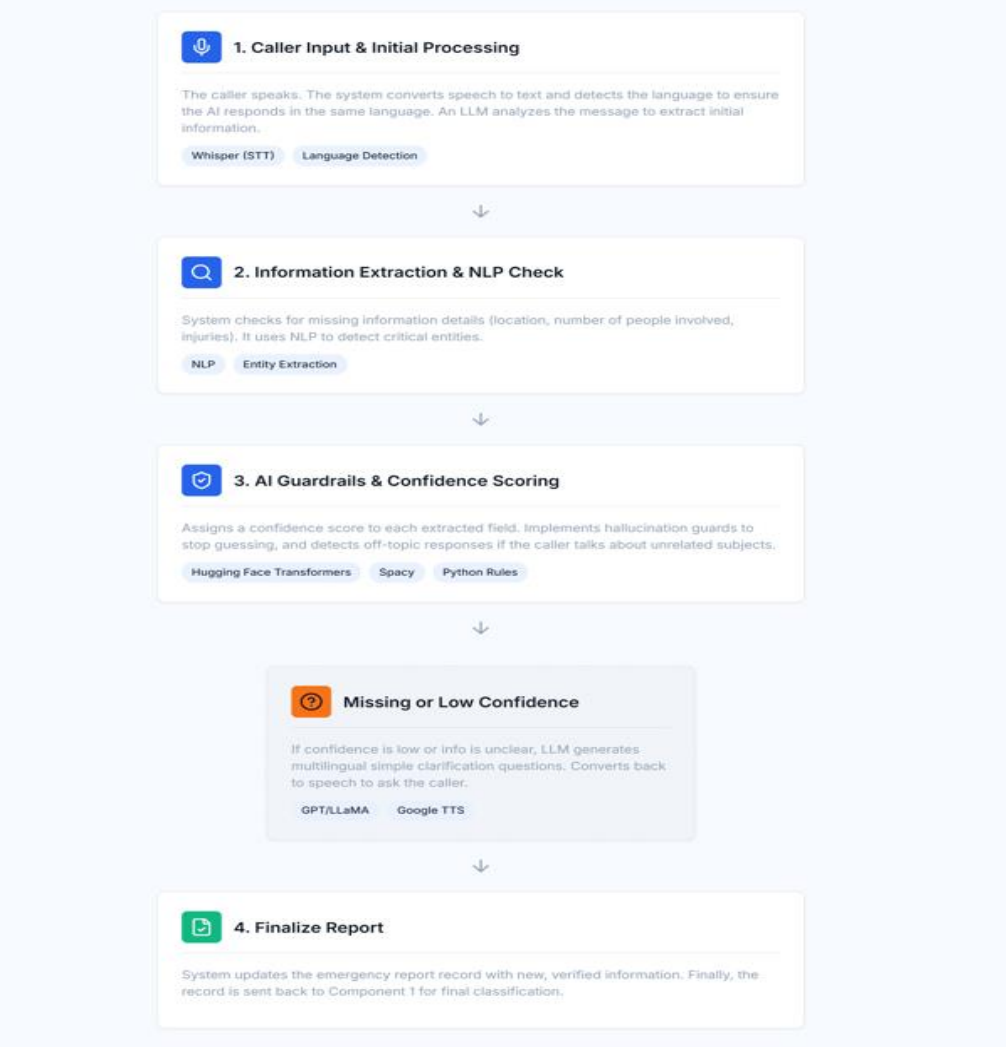


Figure 1: Multilingual Emergency AI Agent for Rapid Clarification

1.2 Literature Review

Several research studies have investigated the application of artificial intelligence and natural language processing techniques to enhance emergency call triage and decision support systems.

He et al. [1] presented an intelligent telephone triage system called **InTriage** that employs natural language processing to analyze conversations between emergency callers and telephone operators. The intelligent telephone triage system has the potential to support emergency call handling through AI-driven analysis.

Similarly, Balaban et al. [2] proposed a multilingual natural language understanding system called **TriagE-NLU**. It is intended for use in emergency dialogue analysis. It emphasizes the need for multilingual support in emergency communication systems and presents an application of AI in supporting triage decisions through the extraction of relevant information from dialogue.

Gormley et al. [3] studied the feasibility of using natural language processing in supporting telephone triage in emergency medical services. It presents a clear case of the potential of using NLP systems in supporting telephone triage decisions through the identification of critical medical symptoms and relevant information from caller dialogue.

Other studies have concentrated on improving the classification of emergency calls using machine learning techniques. Sánchez et al. [4] presented a deep learning-based approach in developing a machine learning model used in the classification of emergency calls. It used a multitask classification model based on the analysis of multimodal data from emergency calls. It presents a clear case of how multiple data sources can be used in improving the performance of emergency dispatch systems.

Emotion detection is another important area of research in emergency communication. Deschamps-Berger et al. [5], [6] investigated the various techniques of speech emotion recognition for emergency call center conversations. The research indicates that the analysis of various acoustic features of the speech of the caller, such as pitch, speaking speed, and vocal intensity, can be effective in the detection of emotional distress in the caller.

Significant advancements have been achieved in the field of conversational AI due to recent developments in large-scale language models. Brown et al. [7] proposed large language models that can perform complex language understanding tasks with minimum supervision. Devlin et al. [8] proposed the BERT model, which achieved significant improvements in the performance of the conversational AI.

In addition, the performance of speech recognition technology like Whisper has been impressive in the transcription of spoken languages into text, even in noisy environments [9]. Dialogue management frameworks using probabilistic models have also been proposed for effective dialogue management in spoken dialogue systems [10].

The studies presented above have shown considerable advancements in the development of AI-based emergency communication systems. However, most of the proposed systems are based on speech recognition technology, call classification, emotion recognition, etc. There are no systems that are particularly designed for interactive AI agents that pose queries to seek missing emergency information.

1.3 Research Gap

However, some limitations are still present in existing emergency triage systems in spite of significant advancements in AI and emergency communication systems.

Firstly, existing emergency systems are mostly used to analyze emergency calls and are not used to interact with callers to gather information that is missing. Although existing systems are able to gather information from conversations using natural language processing and speech recognition techniques, these systems are mostly dependent on human operators to ask questions about incomplete information [1], [3].

Secondly, the emergency dialogue systems that are currently available are mostly intended for the use of high-resource languages such as English. This means that the system may not be effective in a multilingual communication setting, which is common in Sri Lanka [2].

Thirdly, though the emotion detection and speech analysis techniques have been proposed as a means of determining the levels of distress in the emergency callers, they are not often combined with intelligent agents that are capable of dynamically adapting the questioning strategies during emergency calls [5], [6].

Moreover, the emergency AI systems that are currently available cannot prioritize the missing information and generate the necessary questions for the purpose of gathering the necessary information.

Hence, there is a dire need to design a multilingual AI-based conversational agent that can automatically recognize the lack of emergency information and interact with users to obtain critical information in real time. The **proposed Multilingual Emergency AI Agent for Rapid Clarification** is aimed at addressing this problem through the integration of various techniques such as natural language processing and large language models to help emergency operators obtain accurate and complete information in a timely manner.

The proposed solution is aimed at providing a novel solution to improve emergency call triage systems in disaster-prone regions through its focus on multilingual communication, questioning strategies, and information summarization.

2. OBJECTIVE

2.1 Main Objective

For building a multilingual AI agent to quickly identify and clarify any missing critical information from emergency callers in Sinhala, Tamil, English, and code-mixed forms to ensure accurate and complete structured summaries within 30 seconds per call.

2.2 Specific Objectives

- Develop a real-time system for identifying incomplete information from outputs received from Component 1, such as location, headcount, and immediate danger.
- Develop a dynamic question-asking system in multiple languages, responding to caller panic, confusion, and language switching during a call.
- Reduce the involvement of the human operator by ensuring at least 90% completeness of the information related to emergency situations, identified by the AI agent.
- Ensure robustness in dealing with call dropout, unexpected responses, and generating consistent summaries in a specific JSON format.
- Evaluate the efficiency of the AI agent by using parameters such as time-to-complete, completeness rate, satisfaction levels, and adaptability.

3. METHODOLOGY

3.1 Research Approach

The research will adopt a **Design Science** research method, where the emphasis will be on designing and testing the AI agent as a technological solution for addressing the problem of incomplete emergency call information. Iterative design and testing will be adopted as a method for developing the AI agent.

3.2 Technical Methods and Algorithms

1. **Missing Information Detection –** Evaluate structured data from Component 1 (voice intelligence) to detect missing or uncertain data such as location, headcount, and immediate danger.
2. **Dynamic Question Generation** – Leverage Natural Language Generation (NLG) to develop short and context-specific questions based on the caller's language, emotional state, and urgency levels.
3. **Language Adaptation & Code-Mixing Handling –** Integrate an NLP system that supports Sinhala, Tamil, and English languages and also handles code-mixed content using transformer-based architectures (mBERT and XLM-R) pre-trained on Sri Lankan data.
4. **Emotion & Behavior Adaptation** – Integrate rule-based prosody and dialogue management to adapt questions based on the caller's state of mind, whether they're panicked, confused, or distracted.
5. **Structured Summary Generation –** Combine the collected data into an informative and clean JSON-formatted summary for the operators.

3.3 Tools and Platforms with Justification

- **Python –** Primary language for AI and NLP development.
- **Hugging Face Transformers –** Pre-trained multilingual models for NLU and NLG.
- **Speech Recognition Libraries (Whisper / Vosk) –** Integration with Component 1 outputs for speech input handling.
- **Flask / FastAPI** – Backend server for managing real-time interactions between the agent and caller.
- **PostgreSQL / MongoDB –** Database for storing conversation logs, summaries, and evaluation results.
- **Jupyter Notebook / Colab –** For experimentation and training of models.

3.4 Data Collection and Ethical Considerations

- **Data Sources:**
    - Simulated emergency calls from volunteer speakers who provide consented voice recordings and SMS messages.
    - Synthetic data augmentation techniques like text-to-speech conversion, noise injection, and code-mixed text generation.
- **Ethical Considerations:**
    - Summary of Informed Consent from all participants.
    - Anonymization of Informed Consent from all participants.
    - Secure storage of data according to ethical guidelines.
    - Ethical clearance will be sought from the research ethics committee of SLIIT.

3.5 Validation Strategy and Performance Metrics

1. **Time-to-Completion:** The average time taken by the AI agent to generate a complete summary (<30 seconds per call).
2. **Completeness Rate:** Percentage of critical fields filled by the AI agent (target >90%).
3. **Operator Satisfaction:** Measure the usability and clarity of the summaries generated by the AI agent through a survey of the emergency operators.
4. **Robustness:** Test the handling of mid-call language switching, interruptions, and the occurrence of panic.
5. **Accuracy Metrics:**
    - Field extraction accuracy (precision, recall, F1-score).
    - Error rates in the prioritization of questions and follow-ups.
6. **Simulation Testing:** Test the AI agent in a simulated disaster situation.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system is a Multilingual Emergency Conversational AI Agent that is intended to assist in emergency communication during situations of disaster by communicating with callers and gathering missing critical information in emergencies. During emergencies, callers may be under a lot of stress, leading to incomplete or unclear information being given. The proposed system is intended to solve this problem by intelligently communicating with callers and gathering all critical information correctly.

The proposed system is comprised of various connected components that function collectively as a structured pipeline. First, communication with callers in emergencies is achieved through real-time voice communication using Twilio, which receives and streams incoming emergency calls. The speech of the caller is then converted into text using Whisper's Speech-to-Text Technology. The next step is to identify whether the caller is speaking Sinhala, Tamil, English, or a mix of all three using a language detection module.

Next, the text is processed by the Large Language Model, such as GPT or LLaMA, to comprehend the message of the caller and identify key details related to the emergency situation. This step marks the beginning of the creation of the structured emergency report. Subsequently, using natural language processing techniques such as Hugging Face Transformers and spaCy, the system identifies key information such as the location, the number of individuals involved, and the nature of the emergency situation, as well as the confidence level assigned to the extracted information.

Once the initial information is extracted from the caller, the system checks for missing information. If there are any key information gaps, the AI agent creates basic multilingual questions related to the caller's information based on the language used by the caller. These questions are then asked back to the caller, and the answers are converted back to text using the Whisper library.

If the system is unsure of the answers provided, it may ask short questions to ensure that the provided information is correct. In addition, there is an off-topic detection module that checks whether the caller is providing unrelated information. If the information is unrelated, the system uses NLP to redirect the emergency context. There is also a hallucination guard that uses rule-based validation and confidence checks. If the system is unsure of the provided information, it is marked as unknown instead of making assumptions. This cycle continues until all the required information is gathered. Once the required information is gathered, the emergency report is sent to Component 1, which is responsible for the classification and prioritization of emergency cases.

The data flow of the emergency reporting system starts with the voice input of the caller. Then, the input is converted to text using speech-to-text technology. Next, language detection is done, followed by information extraction. Then, there is a check for any missing information. After that, clarification questions are asked, and the information is validated.

Finally, the report is generated and transmitted. All the components work in harmony with each other in order to ensure smooth communication.

From the point of view of scalability, the system is developed in a way that it is capable of serving a large number of emergency calls at the same time, thanks to the cloud-based services that are utilized during the development of the system. On the other hand, the security of the system is addressed by ensuring the privacy of the users, as the system does not store any information about the individuals permanently, utilizing the HTTPS protocol as the communication protocol, as well as addressing the aspect of hallucination control.

Therefore, the system architecture is developed in a way that it is efficient, accurate, and multilingual, allowing the communication between the emergency callers and the emergency services efficiently, thus reducing the time taken during the rescue operations.



Figure 2: HIGH-LEVEL SYSTEM ARCHITECTURE

5. USER REQUIREMENTS

5.1 Stakeholder Identification

1. **Emergency Call Operators –** These are the primary users who will be provided with structured summaries by the AI agent.
2. **Disaster Management Authorities –** These are the supervisors who will be monitoring the efficiency of response and resource allocation.
3. **Civilians / Callers –** These are the indirect users who will be affected by the AI agent, as it will ensure that their information is captured in a timely manner.
4. **First Responders –** These are the indirect users who will be affected by the AI agent, as it will ensure that they are provided with accurate information.
5. **System Administrators / Developers** – These are the users who will be responsible for managing the system's performance.

5.2 Functional Requirements

1. The AI agent will automatically recognize any incomplete or missing critical information from the output of Component 1, which includes location, number of people affected, and type of emergency.
2. The system will develop short and context-specific follow-up questions in the caller's language (Sinhala, Tamil, English, and code-mixed).
3. The AI agent will change the style of questions according to the caller's behavior (e.g., panic, confusion, and interruption).
4. The system will allow the caller to change the language in the middle of the call and will also handle code-mixed responses.
5. The AI agent will develop an entire and structured JSON report of the collected data for the operator's dashboard.
6. The system will be able to handle call interruption and disconnection by developing a partial report and continuing the data collection process.
7. The AI agent will keep track of the questions and responses for further analysis and validation.
8. The system will allow the integration of Component 4 for prioritization and ethical decision-making.

5.3 Non-Functional Requirements

1. **Performance:** The performance of the AI agent should be such that it provides complete summaries in 30 seconds or less per call in normal conditions.
2. **Accuracy:** The completeness rate of critical fields shall be at least 90%, with a low false negative rate.
3. **Usability:** The interface provided to the operator shall clearly and intuitively display the summaries generated by the AI agent.
4. **Reliability:** The AI agent shall provide consistent performance under high call volumes and disaster noise conditions.
5. **Security**: All data shall be encrypted during transmission and at rest, with anonymization of personal data.
6. **Scalability:** The AI agent shall handle multiple calls concurrently without compromising performance.
7. **Resilience:** The AI agent shall function with partial functionality in low bandwidth or network connectivity conditions.
8. **Ethical Compliance:** The AI agent shall not provide misleading or hazardous instructions and shall record all decisions.

6. COMMERCIALIZATION PLAN

The major customer is the Disaster Management Centre (DMC), which is a government agency with the legal responsibility to respond to disasters. The major decision-makers are the Director General of the DMC and the IT and Operations divisions. The system addresses the business need of the DMC to speed up the response time and information overload in the response to disasters. There is also a secondary market of provincial emergency operations centers in the country's 9 provinces. There is also an export market consisting of disaster response agencies in the countries of South Asia, including the Bangladesh NDMA and the India-based National Disaster Response Force (NDRF), which also face the challenge of dealing with multilingual communications in low-resource languages.

The product offers three value propositions that can be quantified. These are: Speed – reducing the time taken to process duplicate reports from 15-20 minutes to less than 8 seconds per event. Safety – ensuring that critical events are not missed due to information overload with the urgency-sensitive safety architecture. Intelligence – enabling DMC operators to make decisions to dispatch resources with the structured intelligence from the incidents instead of raw call records.

The revenue streams align with the Government Technology (GovTech) Software as a Service Licensing Model, which is suitable for the Sri Lanka Public Sector Procurement Environment. The revenue streams are tiered as follows: (1) Years 1-2: Research Pilot with DMC under a sponsored research agreement, with no license fee, funded by NSF Sri Lanka; (2) Years 3-4: national deployment license with DMC at 3.5M LKR per year, including 1 year of hosting, support, and updates; and (3) Years 5+: regional export license with Bangladesh NDMA and India NDRF at 50,000 - 120,000 USD per year per country.

The data insights subscription service offers an additional revenue stream of 1.2M LKR per year at the national level.

The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support gradual adoption, sustainability, and scalability. The pricing strategy for the C4 system is intended to support. In the national deployment phase (Years 3 to 4), the system is planned to be offered as an annual Software as a Service (SaaS) license to DMC and Provincial EOCs in Sri Lanka, priced annually at an estimated LKR 3.5 million, which also includes system maintenance and infrastructure services. In the regional expansion phase (Year 5 and onward), the system is planned to be adapted and offered to international disaster management organizations such as the Bangladesh National Disaster Management Authority and India’s National Disaster Response Force, priced annually between $50,000 and $120,000 as a regional license depending on the scale of deployment and customization required. In addition to the above services offered to DMC and

other research partners, an Analytics Add-on service is planned to be offered to DMC and research partners for post-disaster data analysis and insights reporting, priced annually at approximately LKR 1.2 million.

The competitive advantages of the system are: (1) the first-mover advantage in the multilingual emergency fusion marketplace of South Asia—no competing product speaks to Sinhala or Tamil; (2) the Sri Lanka emergency location gazetteer is a proprietary data asset that cannot be replicated without years of linguistics fieldwork; (3) the open-source technology stack of LaBSE, DBSCAN, PostgreSQL, and React minimizes the cost of licenses; (4) the urgency-sensitive safety architecture is a patentable new design that has yet to be replicated by competitors.

Three IP assets to be developed: (1) Sri Lanka emergency locations gazetteer to be filed as a data publication and issued a suitable Creative Commons license for research use; (2) urgency-sensitive DBSCAN clustering algorithm to be filed as a provisional patent with the National Intellectual Property Office of Sri Lanka (NIPO) after system completion; and (3) research paper to be published to claim scientific priority. The university IP policy and the sponsored research agreement between DMC and SLIIT are applicable.

7. BUDGET AND JUSTIFICATION

Category	Total (LKR)
Software Licenses	60,000
Cloud Services	30,000
Commercialization & IP	20,000
Miscellaneous	10,000
TOTAL 1-YEAR PROJECT BUDGET	**LKR 120000**

Table 1: BUDGET AND JUSTIFICATION

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase / Timeline	Component	Responsible Member	Key Tasks	Workload Contribution
Month 1 – Month 2	Project Planning	All Members	Literature Review, Requirement Analysis, System Architecture Design, Dataset Planning	Shared
Month 3 – Month 5	Voice-Based Language Identification, Incident Understanding & Urgency Detection	Member 1	Audio Preprocessing, Speech-to-Text Transcription, Multilingual Language Detection, Panic/Prosody Analysis, Incident Classification, Location Information, Urgency Detection, Generating Structured JSON Output	Core responsibility of Member 1
Month 3 – Month 5	Multilingual Emergency AI Clarification Agent	Member 2	AI Conversation Agent Design, Information Not Provided, Question Generation, Multilingual Conversation Support (Sinhala, Tamil, English), Panic-Aware Conversation, Handling Interruptions and Call	Core responsibility of Member 2

			Drops, Generating Structured Conversation Summaries	
Month 3 – Month 5	Low-Bandwidth Text/SMS Emergency Intelligence	Member 3	Language Detection for Short Messages, Normalization of Short Message Text, Handling Emojis and Abbreviations, Handling Romanized Sinhala and Tamil, Incident Information, Urgency, Generating Structured SMS Intelligence Output	Core responsibility of Member 3
Month 6 – Month 7	Multimodal Incident Fusion, Timeline Reconstruction & Dashboard	Member 4	Integration of output from components 1-3, incident clustering, duplicate report detection, conflict resolution, creation of a unified incident record, timeline reconstruction, confidence scoring, visualization using a dashboard	Core responsibility of Member 4
Month 8	System Integration	All Members	Integration of all four components,	Shared

			communication via API, testing data flow between modules	
Month 9	System Testing & Evaluation	All Members	Model evaluation, testing, measuring, validation	Shared
Month 10	Dashboard Improvement	Member 4+ Team	Improvement in visualizations, usability testing, refinement	Shared with lead Member 4
Month 11	System Optimization	All Members	Optimization, bug fixes, testing	Shared
Month 12	Documentation & Presentation	All Members	Writing final report, documentation, research paper, final presentation	Shared

Table 2: WORK BREAKDOWN STRUCTURE (WBS)

9. GANTT CHART

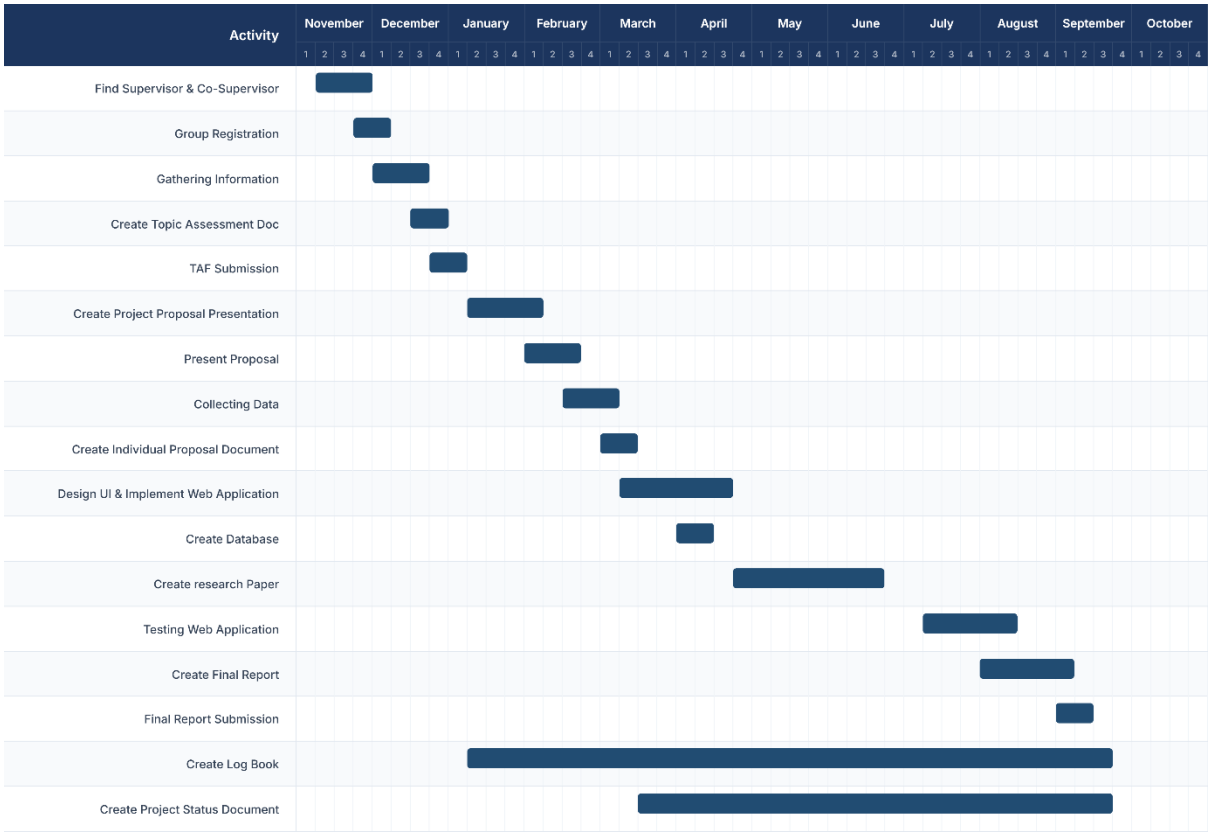


Figure 3: GANTT CHART

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11.APPENDICES

11.1 Preliminary UI Mockups

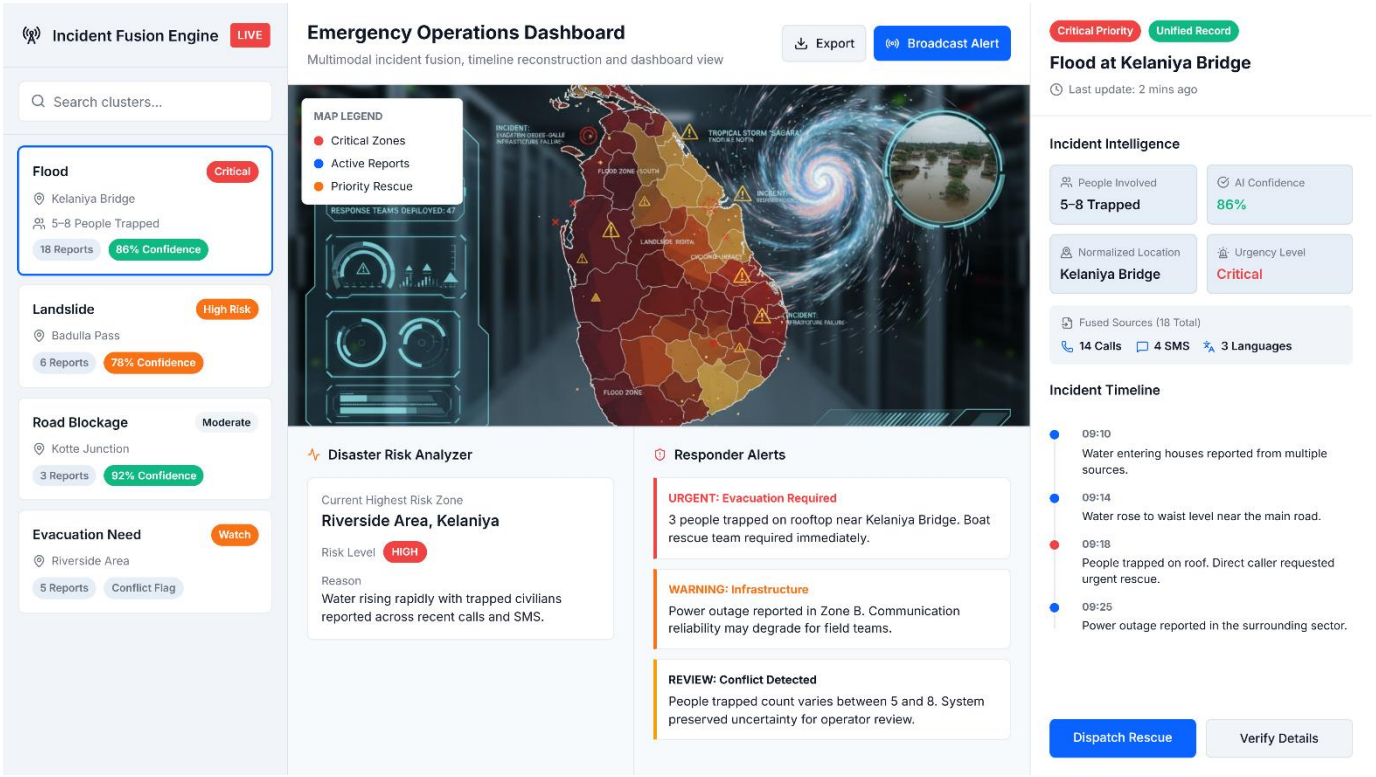


Figure 4: Pre liminary UI Mockups

11.2 Risk Analysis

The importance of risk analysis lies in identifying possible risks that may pose a challenge to the development and operation of the proposed Multilingual Emergency AI System. By identifying these risks, appropriate mitigation measures can be established to minimize their effect on the system.

1. Technical Risks

Technical risks are likely to occur in the system in the following ways: The system may fail to identify the language of the caller or fail to understand the information conveyed by the caller due to poor call quality. This risk can be minimized by ensuring that the system uses good models in its operation and improving the models to match the updated data.

2. Operational Risks

Operational risks happen during real-time emergency interactions. Callers might panic and provide unclear information or hang up the call unexpectedly. This might affect the emergency agent's ability to gather clear emergency information from callers. To address this risk, the system will be designed to include an adaptive conversation manager to clarify questions and an interruption handler to resume the conversation in case the call is re-established.

3. Data Privacy and Security Risks

The system handles sensitive emergency information such as caller locations and personal details. Unauthorized access to emergency data might compromise the privacy of callers. To address this risk, the system will be designed to include security features to protect sensitive emergency data from unauthorized access.

4. Scalability Risks

In the case of a disaster or an emergency, the number of incoming calls to the system may increase, and this is a risk to the system. This risk can be managed by implementing a scalable system that can handle the workload.

5. Integration Risks

The system has to integrate with other systems that are used in emergencies and communication systems. This risk can be managed by implementing a system that can integrate with other systems to ensure the efficiency of the system.